

Kubernetes Day 3

- Gitbash

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/MC-B17D/Kubernetes
$ aws --version
aws-cli/2.34.53 Python/3.14.5 windows/11 exe/AMD64
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/MC-B17D/Kubernetes
$ eksctl version
0.226.0
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/MC-B17D/Kubernetes
$ kubectl version --client
Client Version: v1.35.3-eks-bbe087e
Kustomize Version: v5.7.1
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/MC-B17D/Kubernetes
$ kubectl version --client
Client Version: v1.35.3-eks-bbe087e
Kustomize Version: v5.7.1
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/MC-B17D/Kubernetes
$ eksctl create cluster --name mccluster --nodegroup-name b17dng --node-type t3.micro --nodes 5
--managed
```

```
MINGW64/C:/Users/Bhavana/OneDrive/Desktop
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ eksctl create cluster --name mccluster --nodegroup-name b17dng --node-type t3.micro --nodes 5 --managed
2026-06-01 12:51:31 [i] eksctl version 0.226.0
2026-06-01 12:51:31 [i] using region us-east-1
2026-06-01 12:51:33 [i] setting availability zones to [us-east-1c us-east-1f]
2026-06-01 12:51:33 [i] subnets for us-east-1c - public:192.168.0.0/19 private:192.168.64.0/19
2026-06-01 12:51:33 [i] subnets for us-east-1f - public:192.168.32.0/19 private:192.168.96.0/19
2026-06-01 12:51:33 [i] nodegroup "b17dng" will use "" [AmazonLinux2023/1.34]
2026-06-01 12:51:33 [i] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To opt-out of this behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-06-01 12:51:33 [i] using Kubernetes version 1.34
2026-06-01 12:51:33 [i] creating EKS cluster "mccluster" in "us-east-1" region with managed nodes
2026-06-01 12:51:33 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-06-01 12:51:33 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=mccluster'
2026-06-01 12:51:33 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "mccluster" in "us-east-1"
2026-06-01 12:51:33 [i] Cloudwatch logging will not be enabled for cluster "mccluster" in "us-east-1"
2026-06-01 12:51:33 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=mccluster'
2026-06-01 12:51:33 [i] default add-ons kube-proxy, coredns, metrics-server, vpc-cni were not specified, will install them as EKS add-ons
2026-06-01 12:51:33 [i]
2 sequential tasks: { create cluster control plane "mccluster",
2 sequential sub-tasks: {
2 sequential sub-tasks: {
1 task: { create add-ons },
wait for control plane to become ready,
},
create managed nodegroup "b17dng",
}
}
2026-06-01 12:51:33 [i] building cluster stack "eksctl-mccluster-cluster"
2026-06-01 12:51:34 [i] deploying stack "eksctl-mccluster-cluster"
2026-06-01 12:52:04 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:52:35 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:53:36 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:54:37 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:55:39 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:56:40 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:57:41 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:58:43 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:59:44 [i] waiting for CloudFormation stack "eksctl-mccluster-cluster"
2026-06-01 12:59:49 [i] creating add-on: kube-proxy
2026-06-01 12:59:50 [i] successfully created add-on: kube-proxy
2026-06-01 12:59:51 [i] creating add-on: coredns
2026-06-01 12:59:53 [i] successfully created add-on: coredns
2026-06-01 12:59:54 [i] recommended policies were found for "vpc-cni" add-on, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended permissions for "vpc-cni" add-on is via pod identity associations; after add-on creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and add-on
2026-06-01 12:59:54 [i] creating add-on: vpc-cni
2026-06-01 12:59:54 [i] successfully created add-on: vpc-cni
2026-06-01 13:01:59 [i] building managed nodegroup stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 13:02:00 [i] deploying stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 13:02:00 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
```

```
2026-06-01 12:59:54 [i] recommended policies were found for "vpc-cni" add-on, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended permissions for "vpc-cni" add-on is via pod identity associations; after add-on creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and add-on
2026-06-01 12:59:54 [i] creating add-on: vpc-cni
2026-06-01 12:59:54 [i] successfully created add-on: vpc-cni
2026-06-01 13:01:59 [i] building managed nodegroup stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 13:02:00 [i] deploying stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 13:02:00 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 13:02:32 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 13:03:26 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 13:04:46 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 13:04:47 [i] waiting for the control plane to become ready
2026-06-01 13:04:47 [i] saved kubeconfig as "C:\Users\Bhavana\.kube\config"
2026-06-01 13:04:47 [i] no tasks
2026-06-01 13:04:47 [i] all EKS cluster resources for "mccluster" have been created
2026-06-01 13:04:48 [i] nodegroup "b17dng" has 5 node(s)
2026-06-01 13:04:48 [i] node "ip-192-168-27-140.ec2.internal" is ready
2026-06-01 13:04:48 [i] node "ip-192-168-28-100.ec2.internal" is ready
2026-06-01 13:04:48 [i] node "ip-192-168-3-163.ec2.internal" is ready
2026-06-01 13:04:48 [i] node "ip-192-168-37-206.ec2.internal" is ready
2026-06-01 13:04:48 [i] node "ip-192-168-58-58.ec2.internal" is ready
2026-06-01 13:04:48 [i] waiting for at least 5 node(s) to become ready in "b17dng"
2026-06-01 13:04:49 [i] nodegroup "b17dng" has 5 node(s)
2026-06-01 13:04:49 [i] node "ip-192-168-27-140.ec2.internal" is ready
2026-06-01 13:04:49 [i] node "ip-192-168-28-100.ec2.internal" is ready
2026-06-01 13:04:49 [i] node "ip-192-168-3-163.ec2.internal" is ready
2026-06-01 13:04:49 [i] node "ip-192-168-37-206.ec2.internal" is ready
2026-06-01 13:04:49 [i] node "ip-192-168-58-58.ec2.internal" is ready
2026-06-01 13:04:49 [i] created 1 managed nodegroup(s) in cluster "mccluster"
2026-06-01 13:04:49 [i] creating add-on: metrics-server
2026-06-01 13:04:51 [i] successfully created add-on: metrics-server
2026-06-01 13:04:53 [i] kubectl command should work with "C:\Users\Bhavana\.kube\config", try 'kubectl get nodes'
2026-06-01 13:04:53 [i] EKS cluster "mccluster" in "us-east-1" region is ready
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
```

```
$ eksctl get cluster
```

NAME	REGION	EKSCTL CREATED
mccluster	us-east-1	True

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
```

```
$ kubectl cluster-info
```

Kubernetes control plane is running at https://9A5B546A5CCDDDF689A02A80F6D5A6BC.gr7.us-east-1.eks.amazonaws.com

CoreDNS is running at https://9A5B546A5CCDDDF689A02A80F6D5A6BC.gr7.us-east-1.eks.amazonaws.com/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
```

```
$ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
ip-192-168-27-140.ec2.internal	Ready	<none>	8m43s	v1.34.8-eks-3385e9b
ip-192-168-28-100.ec2.internal	Ready	<none>	8m43s	v1.34.8-eks-3385e9b
ip-192-168-3-163.ec2.internal	Ready	<none>	8m43s	v1.34.8-eks-3385e9b
ip-192-168-37-206.ec2.internal	Ready	<none>	8m41s	v1.34.8-eks-3385e9b
ip-192-168-58-58.ec2.internal	Ready	<none>	8m45s	v1.34.8-eks-3385e9b

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
```

```
$ kubectl get pods
```

No resources found in default namespace.

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
```

```
$ kubectl run pod1 --image tomcat
```

pod/pod1 created

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
```

```
$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
pod1	0/1	ContainerCreating	0	7s

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
```

```
$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	12s

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
```

```
$ kubectl describe pod pod1
```

```
Name: pod1
Namespace: default
Priority: 0
Service Account: default
Node: ip-192-168-3-163.ec2.internal/192.168.3.163
Start Time: Mon, 01 Jun 2026 13:26:39 +0530
Labels: run=pod1
Annotations: <none>
Status: Running
IP: 192.168.0.248
IPs:
  IP: 192.168.0.248
Containers:
  pod1:
    Container ID: containerd://5306d92c67f0ed167b4cb07902faf3954ad87bf7052a66c71f3c2ded078074f4
    Image: tomcat
    Image ID: docker.io/library/tomcat@sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
    Port: <none>
    Host Port: <none>
    State: Running
      Started: Mon, 01 Jun 2026 13:26:46 +0530
    Ready: True
    Restart Count: 0
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-2rsr4 (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                       True
  Ready                             True
  ContainersReady                   True
  PodScheduled                      True
Volumes:
  kube-api-access-2rsr4:
    Type: Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    Optional: false
    DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
              node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
```

Events:	Type	Reason	Age	From	Message
	Normal	Scheduled	41s	default-scheduler	Successfully assigned default/pod1 to ip-192-168-3-163.ec2.internal
	Normal	Pulling	40s	kubelet	spec.containers{pod1}: Pulling image "tomcat"
	Normal	Pulled	34s	kubelet	spec.containers{pod1}: Successfully pulled image "tomcat" in 5.754s (5.754s including waiting). Image size: 154490448 bytes.
	Normal	Created	34s	kubelet	spec.containers{pod1}: Created container: pod1
	Normal	Started	34s	kubelet	spec.containers{pod1}: Started container pod1

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl run pod2 --image tomiiimcat
pod/pod2 created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	4m7s
pod2	0/1	ErrImagePull	0	11s

(or) ImagePullBackoff

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl describe pod pod2

```

Name: pod2
Namespace: default
Priority: 0
Service Account: default
Node: ip-192-168-28-100.ec2.internal/192.168.28.100
Start Time: Mon, 01 Jun 2026 13:30:36 +0530
Labels: run=pod2
Annotations: <none>
Status: Pending
IP: 192.168.28.228
IPs:
  IP: 192.168.28.228
Containers:
  pod2:
    Container ID:
    Image: tomiiimcat
    Image ID:
    Port: <none>
    Host Port: <none>
    State: Waiting
      Reason: ImagePullBackOff
    Ready: False
    Restart Count: 0
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-hbtmq (ro)

```

```

Conditions:
  Type                               Status
  PodReadyToStartContainers          True
  Initialized                         True
  Ready                              False
  ContainersReady                    False
  PodScheduled                       True
Volumes:
  kube-api-access-hbtmq:
    Type: Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    Optional: false
    DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
              node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

```

Events:	Type	Reason	Age	From	Message
	Normal	Scheduled	2m6s	default-scheduler	Successfully assigned default/pod2 to ip-192-168-28-100.ec2.internal
	Normal	Pulling	47s (x4 over 2m6s)	kubelet	spec.containers{pod2}: Pulling image "tomiiimcat"
	Warning	Failed	47s (x4 over 2m6s)	kubelet	spec.containers{pod2}: Failed to pull image "tomiiimcat": failed to pull and unpack image "docker.io/library/tomiiimcat:latest": failed to resolve reference "docker.io/library/tomiiimcat:latest": pull access denied, repository does not exist or may require authorization: server message: insufficient_scope: authorization failed
	Warning	Failed	47s (x4 over 2m6s)	kubelet	spec.containers{pod2}: Error: ErrImagePull
	Normal	BackOff	12s (x7 over 2m5s)	kubelet	spec.containers{pod2}: Back-off pulling image "tomiiimcat"
	Warning	Failed	12s (x7 over 2m5s)	kubelet	spec.containers{pod2}: Error: ImagePullBackOff

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl run pod3 --image amazonlinux
pod/pod3 created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	9m8s
pod2	0/1	ImagePullBackOff	0	5m12s
pod3	0/1	CrashLoopBackOff	1 (5s ago)	11s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl describe pod pod3

Name: pod3
Namespace: default
Priority: 0
Service Account: default
Node: ip-192-168-3-163.ec2.internal/192.168.3.163
Start Time: Mon, 01 Jun 2026 13:35:36 +0530
Labels: run=pod3
Annotations: <none>
Status: Running
IP: 192.168.22.2
IPs:
IP: 192.168.22.2
Containers:
pod3:
Container ID: containerd://85940d0beacde4887710763ce76ba4cbf148b13aa3d29959cbf9dc8e6531bccf
Image: amazonlinux
Image ID: docker.io/library/amazonlinux@sha256:ad882807c6c88f478186fa2632637380e06071a303e2b3bd96ab71846a77c6a3
Port: <none>
Host Port: <none>
State: Waiting
Reason: CrashLoopBackOff
Last State: Terminated
Reason: Completed
Exit Code: 0
Started: Mon, 01 Jun 2026 13:36:24 +0530
Finished: Mon, 01 Jun 2026 13:36:24 +0530
Ready: False
Restart Count: 3
Environment: <none>
Mounts:
/var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-knv4s (ro)
Conditions:
Type Status
PodReadyToStartContainers True
Initialized True
Ready False
ContainersReady False
PodScheduled True
Volumes:
kube-api-access-knv4s:
Type: Projected (a volume that contains injected data from multiple sources)
TokenExpirationSeconds: 3607
ConfigMapName: kube-root-ca.crt
Optional: false
DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
Type Reason Age From Message

Normal Scheduled 84s default-scheduler Successfully assigned default/pod3 to ip-192-168-3-163.ec2.internal
Normal Pulled 79s kubelet spec.containers{pod3}: Successfully pulled image "amazonlinux" in 3.328s (3.328s including waiting). Image size: 54575140 bytes.
Normal Pulled 78s kubelet spec.containers{pod3}: Successfully pulled image "amazonlinux" in 121ms (121ms including waiting). Image size: 54575140 bytes.
Normal Pulled 62s kubelet spec.containers{pod3}: Successfully pulled image "amazonlinux" in 125ms (125ms including waiting). Image size: 54575140 bytes.
Normal Pulling 36s (x4 over 83s) kubelet spec.containers{pod3}: Pulling image "amazonlinux"
Normal Created 36s (x4 over 79s) kubelet spec.containers{pod3}: Created container: pod3
Normal Started 36s (x4 over 79s) kubelet spec.containers{pod3}: Started container pod3
Normal Pulled 36s kubelet spec.containers{pod3}: Successfully pulled image "amazonlinux" in 152ms (152ms including waiting). Image size: 54575140 bytes.
Warning BackOff 0s (x8 over 78s) kubelet spec.containers{pod3}: Back-off restarting failed container pod3 in pod pod3_default(8d1d5766-8434-4d86-a42a-58c6ad7bd8d0)

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	12m
pod2	0/1	ImagePullBackOff	0	8m21s
pod3	0/1	Completed	5 (102s ago)	3m20s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl delete pod pod2 pod3

pod "pod2" deleted from default namespace
pod "pod3" deleted from default namespace

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	19m

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl run pod4 --image tomcat
pod/pod4 created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	19m
pod4	0/1	ContainerCreating	0	7s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	20m
pod4	1/1	Running	0	11s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl run pod5 --image tomcat
pod/pod5 created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	20m
pod4	1/1	Running	0	52s
pod5	0/1	ContainerCreating	0	4s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	20m
pod4	1/1	Running	0	56s
pod5	1/1	Running	0	8s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl run pod2 --image tomcat
pod/pod2 created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	21m
pod2	1/1	Running	0	3s
pod4	1/1	Running	0	76s
pod5	1/1	Running	0	28s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl run pod3 --image tomcat
pod/pod3 created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	21m
pod2	1/1	Running	0	21s
pod3	1/1	Running	0	3s
pod4	1/1	Running	0	94s
pod5	1/1	Running	0	46s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl run pod6 --image tomcat
pod/pod6 created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	21m
pod2	1/1	Running	0	37s
pod3	1/1	Running	0	19s
pod4	1/1	Running	0	110s
pod5	1/1	Running	0	62s
pod6	0/1	ContainerCreating	0	4s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	21m
pod2	1/1	Running	0	43s

```
pod3 1/1 Running 0 25s
pod4 1/1 Running 0 116s
pod5 1/1 Running 0 68s
pod6 1/1 Running 0 10s
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl run pod7 --image tomcat
 pod/pod7 created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl get po

```
NAME    READY   STATUS    RESTARTS   AGE
pod1    1/1     Running   0          22m
pod2    1/1     Running   0          66s
pod3    1/1     Running   0          48s
pod4    1/1     Running   0          2m19s
pod5    1/1     Running   0          91s
pod6    1/1     Running   0          33s
pod7    0/1     Pending   0          4s
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl describe pod pod7

```
Name:          pod7
Namespace:     default
Priority:       0
Service Account: default
Node:          <none>
Labels:        run=pod7
Annotations:    <none>
Status:        Pending
IP:
IPs:           <none>
Containers:
  pod7:
    Image:      tomcat
    Port:       <none>
    Host Port:  <none>
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-pztdx (ro)
Conditions:
  Type             Status
  PodScheduled     False
Volumes:
  kube-api-access-pztdx:
    Type:          Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    Optional:      false
    DownwardAPI:   true
QoS Class:        BestEffort
Node-Selectors:    <none>
Tolerations:      node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                  node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason             Age    From                      Message
  ----    -
  Warning FailedScheduling   61s    default-scheduler        0/5 nodes are available: 5 Too many pods. no new
claims to deallocate, preemption: 0/5 nodes are available: 5 No preemption victims found for incoming pod.
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl delete pod pod6
 pod "pod6" deleted from default namespace

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl get po

```
NAME    READY   STATUS             RESTARTS   AGE
pod1    1/1     Running            0          24m
pod2    1/1     Running            0          3m37s
pod3    1/1     Running            0          3m19s
pod4    1/1     Running            0          4m50s
pod5    1/1     Running            0          4m2s
pod7    0/1     ContainerCreating  0          2m35s
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl get po

```
NAME    READY   STATUS             RESTARTS   AGE
pod1    1/1     Running            0          24m
pod2    1/1     Running            0          3m41s
pod3    1/1     Running            0          3m23s
pod4    1/1     Running            0          4m54s
pod5    1/1     Running            0          4m6s
pod7    0/1     ContainerCreating  0          2m39s
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl get po

```
NAME    READY   STATUS    RESTARTS   AGE
pod1    1/1     Running   0          24m
```


pod2	1/1	Running	0	3m50s
pod3	1/1	Running	0	3m32s
pod4	1/1	Running	0	5m3s
pod5	1/1	Running	0	4m15s
pod7	0/1	ContainerCreating	0	2m48s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	24m
pod2	1/1	Running	0	3m55s
pod3	1/1	Running	0	3m37s
pod4	1/1	Running	0	5m8s
pod5	1/1	Running	0	4m20s
pod7	0/1	ContainerCreating	0	2m53s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	25m
pod2	1/1	Running	0	4m8s
pod3	1/1	Running	0	3m50s
pod4	1/1	Running	0	5m21s
pod5	1/1	Running	0	4m33s
pod7	0/1	ContainerCreating	0	3m6s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl delete pod pod5

pod "pod5" deleted from default namespace

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	25m
pod2	1/1	Running	0	4m40s
pod3	1/1	Running	0	4m22s
pod4	1/1	Running	0	5m53s
pod7	1/1	Running	0	3m38s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po -o wide

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE
pod1	1/1	Running	0	27m	192.168.0.248	ip-192-168-3-163.ec2.internal	<none>
pod2	1/1	Running	0	5m58s	192.168.8.63	ip-192-168-28-100.ec2.internal	<none>
pod3	1/1	Running	0	5m40s	192.168.22.2	ip-192-168-3-163.ec2.internal	<none>
pod4	1/1	Running	0	7m11s	192.168.28.228	ip-192-168-28-100.ec2.internal	<none>
pod7	1/1	Running	0	4m56s	192.168.33.43	ip-192-168-37-206.ec2.internal	<none>

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get nodes -o wide

NAME	STATUS	ROLES	AGE	VERSION	INTERNAL-IP	EXTERNAL-IP	OS-IMAGE
KERNEL-VERSION	CONTAINER-RUNTIME						
ip-192-168-27-140.ec2.internal	Ready	<none>	51m	v1.34.8-eks-3385e9b	192.168.27.140	52.23.165.83	Amazon
Linux 2023.11.20260526	6.12.88-119.157.amzn2023.x86_64	containerd://2.2.3+unknown					
ip-192-168-28-100.ec2.internal	Ready	<none>	51m	v1.34.8-eks-3385e9b	192.168.28.100	3.90.68.136	Amazon
Linux 2023.11.20260526	6.12.88-119.157.amzn2023.x86_64	containerd://2.2.3+unknown					
ip-192-168-3-163.ec2.internal	Ready	<none>	51m	v1.34.8-eks-3385e9b	192.168.3.163	54.226.66.146	Amazon
Linux 2023.11.20260526	6.12.88-119.157.amzn2023.x86_64	containerd://2.2.3+unknown					
ip-192-168-37-206.ec2.internal	Ready	<none>	51m	v1.34.8-eks-3385e9b	192.168.37.206	3.236.65.49	Amazon
Linux 2023.11.20260526	6.12.88-119.157.amzn2023.x86_64	containerd://2.2.3+unknown					
ip-192-168-58-58.ec2.internal	Ready	<none>	51m	v1.34.8-eks-3385e9b	192.168.58.58	13.223.234.152	Amazon
Linux 2023.11.20260526	6.12.88-119.157.amzn2023.x86_64	containerd://2.2.3+unknown					

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl delete pod1 pod2 pod3 pod4

error: the server doesn't have a resource type "pod1"

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl delete pod pod1 pod2 pod3 pod4

pod "pod1" deleted from default namespace
pod "pod2" deleted from default namespace
pod "pod3" deleted from default namespace
pod "pod4" deleted from default namespace

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
pod7	1/1	Running	0	15m

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes

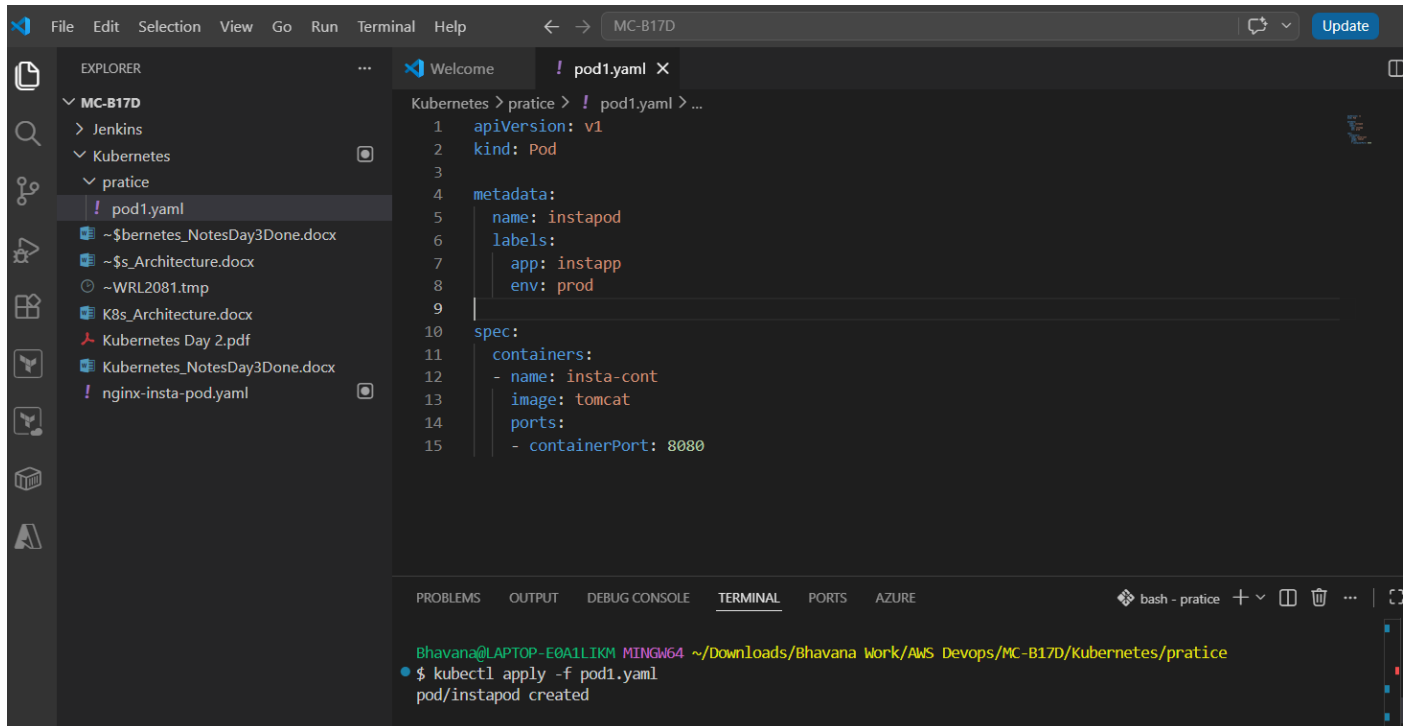
\$ kubectl delete pod pod7

pod "pod7" deleted from default namespace

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
$ kubectl get po
No resources found in default namespace.
```

Manifest Files -- .yaml

- But, In real-time we write manifest files in yaml format to create k8s objects/resources



```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
$ kubectl apply -f pod1.yaml
pod/instapod created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
$ kubectl get po
NAME      READY   STATUS    RESTARTS   AGE
instapod  1/1     Running   0           2m4s
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
$ kubectl describe pod instapod
```

```
Name:      instapod
Namespace: default
Priority:   0
Service Account: default
Node:      ip-192-168-28-100.ec2.internal/192.168.28.100
Start Time: Mon, 01 Jun 2026 14:40:52 +0530
Labels:    app=instapp
           env=prod
Annotations: <none>
Status:    Running
IP:        192.168.28.228
IPs:
  IP: 192.168.28.228
Containers:
  insta-cont:
    Container ID: containerd://d8a8c0235fffb141465a322a3090d5c108d56b1b18911a38ef1fbfb35c999f91f
    Image:        tomcat
    Image ID:     docker.io/library/tomcat@sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
    Port:        8080/TCP
    Host Port:   0/TCP
    State:       Running
      Started:   Mon, 01 Jun 2026 14:40:53 +0530
    Ready:       True
    Restart Count: 0
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-54zh1 (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                       True
  Ready                             True
  ContainersReady                   True
  PodScheduled                      True
Volumes:
```



```

kube-api-access-54zh1:
  Type: Projected (a volume that contains injected data from multiple sources)
  TokenExpirationSeconds: 3607
  ConfigMapName: kube-root-ca.crt
  Optional: false
  DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
              node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

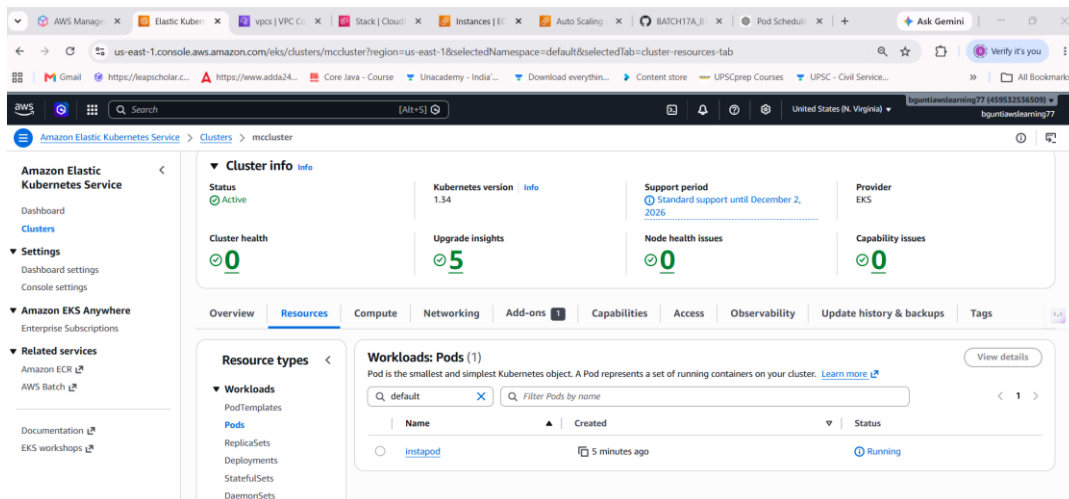
Events:
  Type      Reason      Age   From          Message
  ----      -
Normal    Scheduled   2m25s default-scheduler Successfully assigned default/instapod to ip-192-168-28-100.ec2.internal
Normal    Pulling     2m24s kubelet       spec.containers{insta-cont}: Pulling image "tomcat"
Normal    Pulled      2m24s kubelet       spec.containers{insta-cont}: Successfully pulled image "tomcat" in 150ms (150ms including waiting). Image size: 154490448 bytes.
Normal    Created     2m24s kubelet       spec.containers{insta-cont}: Created container: insta-cont
Normal    Started     2m24s kubelet       spec.containers{insta-cont}: Started container insta-cont

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl logs instapod

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
instapod	1/1	Running	0	4m40s



Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ eksctl get cluster

NAME	REGION	EKSCTL CREATED
mccluster	us-east-1	True

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes
\$ eksctl delete cluster mccluster

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes/pratice
$ eksctl get cluster
NAME      REGION  EKSCTL CREATED
mccluster us-east-1 True

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes/pratice
$ eksctl delete cluster mccluster
2026-06-01 14:49:54 [i] deleting EKS cluster "mccluster"
2026-06-01 14:49:56 [i] will drain 0 unmanaged nodegroup(s) in cluster "mccluster"
2026-06-01 14:49:56 [i] starting parallel draining, max in-flight of 1
2026-06-01 14:49:58 [i] deleted 0 Fargate profile(s)
2026-06-01 14:50:01 [✓] kubeconfig has been updated
2026-06-01 14:50:02 [i] cleaning up AWS load balancers created by Kubernetes objects of Kind Service, Ingress, or Gateway
2026-06-01 14:50:09 [i]
2 sequential tasks: { delete nodegroup "b17dng", delete cluster control plane "mccluster" [async]
}
2026-06-01 14:50:10 [i] will delete stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:50:10 [i] waiting for stack "eksctl-mccluster-nodegroup-b17dng" to get deleted
2026-06-01 14:50:10 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:50:41 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:51:30 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:52:49 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:53:34 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:54:22 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:55:44 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:56:24 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:57:14 [i] waiting for CloudFormation stack "eksctl-mccluster-nodegroup-b17dng"
2026-06-01 14:57:15 [i] will delete stack "eksctl-mccluster-cluster"
2026-06-01 14:57:16 [✓] all cluster resources were deleted

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/MC-B17D/Kubernetes/pratice
$

```